

Configuration and validation of Modbus TCP communication

TSX H57 44M · ETY4103 · TeSys T LTMR08EFM



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I/O Scanning configuration, motor register readings, network diagnostics and validation of the Modbus TCP service.

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1 Abstract

During my internship at the OCP Jorf Lasfar industrial complex, I integrated a Schneider TSX H57 44M PLC with a TeSys T LTMR08EFM motor management controller over Modbus TCP. I configured the TSX ETY4103 Ethernet module in EcoStruxure Control Expert, set up I/O Scanning and mapped the motor controller's status and measurement registers into PLC memory.

The ETY4103 uses address 192.168.8.21 and the TeSys T uses 192.168.8.54, with subnet mask 255.255.255.0, Modbus TCP port 502 and Unit ID 1. I configured two read areas: registers 2502–2505 map to %MW100–%MW103, and registers 500–507 map to %MW104–%MW111. Both write lengths are zero.

Network tests returned successful ping replies from both devices, and TCP port 502 on the TeSys T was reachable. During the low-current test, the displayed average current was 0.13 A and the internal controller temperature was approximately 44 °C. The displayed current is below the LTMR08EFM's stated measurement range (0.4–8 A), so it cannot establish current-measurement accuracy.

2 Background and objectives

TeSys T units combine motor feeder protection, local control and measurement. The goal of this internship project was to connect the LTMR08EFM to a Schneider Premium PLC and make its data available in PLC memory for diagnostics and future supervision.

The technical objectives were to:

- configure the Premium rack in Control Expert with the references matching the installed hardware;
- configure the ETY4103 Ethernet interface;
- establish Modbus TCP communication with the TeSys T;
- map the read registers into PLC %MW words;
- verify IP connectivity and availability of the Modbus TCP service;
- prepare the values for viewing in an animation table.

The implemented I/O Scanner entries acquire data from the motor controller. Remote motor commands over Modbus were outside the scope of this work.

3 Hardware and architecture

Equipment	Reference or configuration	Function
Rack power supply	TSX PSY2600	Power supply for the Premium rack
PLC	TSX H57 44M, version 02.50	Executes the PLC program and stores values in %MW words
Ethernet module	TSX ETY4103, firmware SV 04.80	Ethernet interface and I/O Scanning
Motor controller	TeSys T LTMR08EFM	Motor feeder protection, local control and measurements
Network	Unmanaged Ethernet switch	Connects the PC, ETY4103 and TeSys T
Software	EcoStruxure Control Expert	Hardware and network configuration, transfer and diagnostics

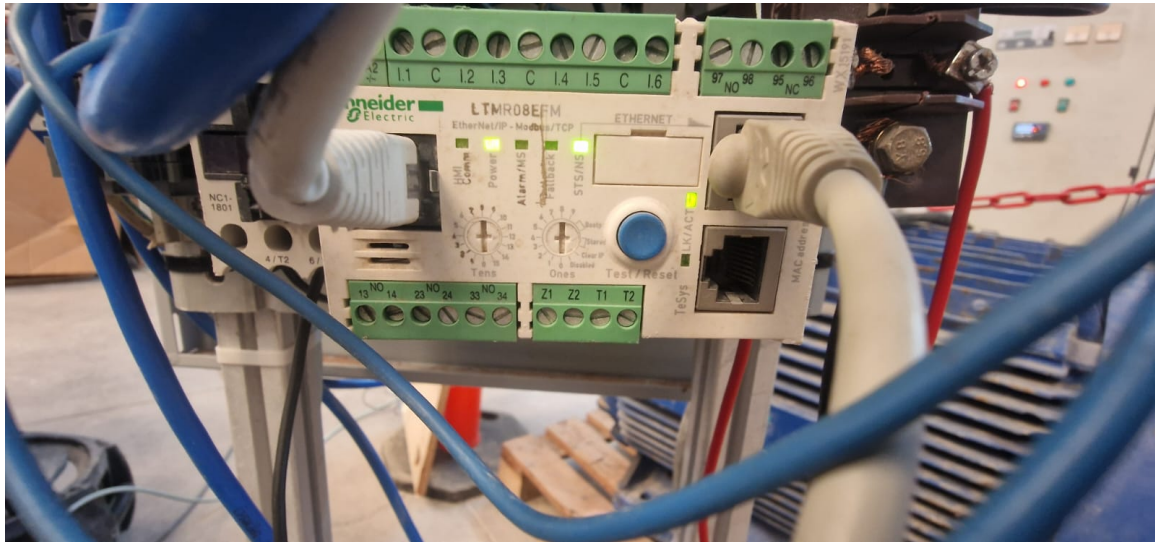


Figure 1: TeSys T motor management controller

The configuration PC connects to the CPU over USB for programming and to the Ethernet network for diagnostics. The USB cable does not carry Modbus TCP traffic. The ETY4103 and TeSys T connect to the switch and, during network testing, share a subnet with the PC Ethernet interface.



Figure 2: Installed control cabinet

4 IP addressing plan

Node	Address	Subnet mask	Purpose
Configuration PC	192.168.8.x (test addresses .12 and .20)	255.255.255.0	Programming and network diagnostics
TSX ETY4103	192.168.8.21	255.255.255.0	I/O Scanning and Modbus TCP client
TeSys T LTMR08EFM	192.168.8.54	255.255.255.0	Modbus TCP server, port 502

The gateway is set to 0.0.0.0 for this isolated local network. Each device must have a unique address on the subnet.

5 Hardware and Ethernet configuration

In Control Expert, I configured the Premium rack with the TSX PSY2600 power supply, TSX H57 44M CPU and ETY4103 in slot 2. I selected CPU catalog version 02.50 to match the installed hardware.

I assigned Configured addressing to the network `eth_tsys`, set IP address 192.168.8.21 and mask 255.255.255.0, and enabled I/O Scanning. Selecting CPU version 03.10 instead of the installed 02.50 led to a compatibility check failure at transfer; matching the version resolved that mismatch.

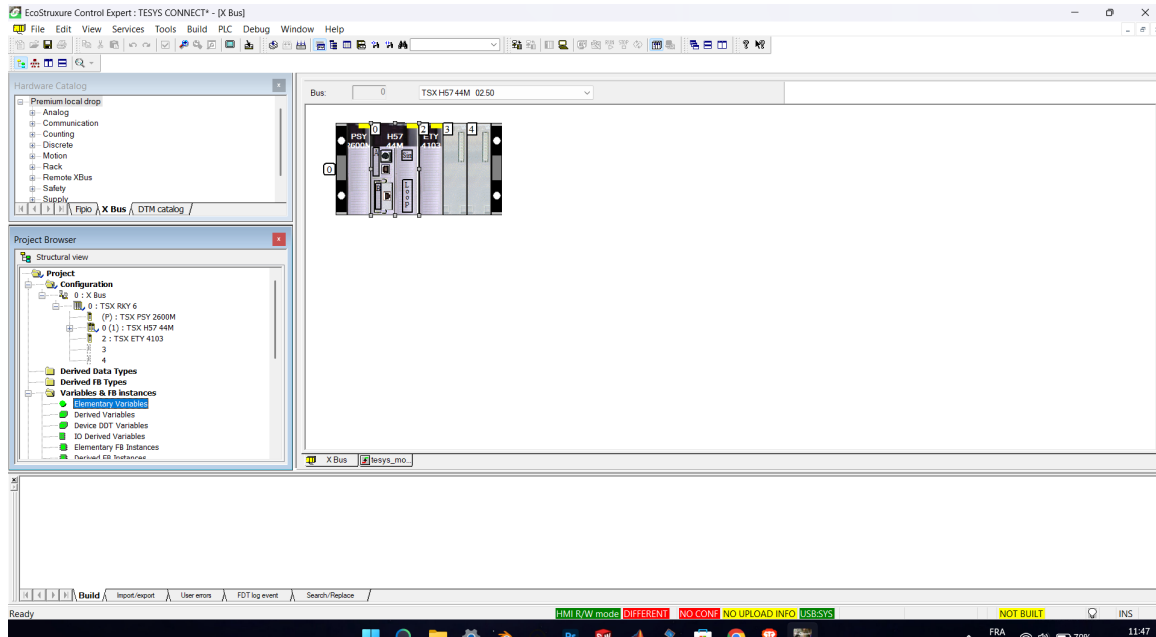


Figure 3: Rack configuration in Control Expert

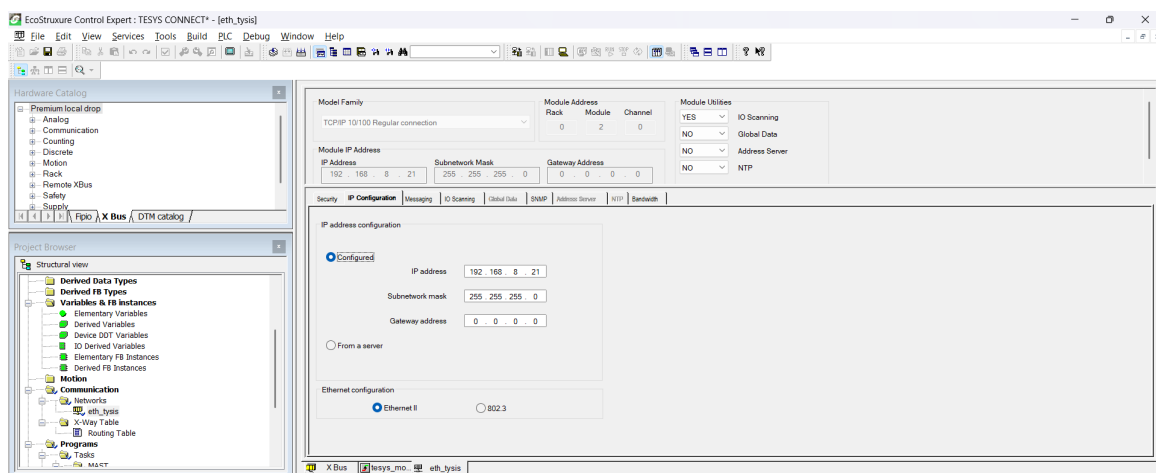


Figure 4: ETY4103 IP configuration

5.1 Service settings

During the test, **Firmware Upgrade & FDR (FTP/TFTP)** and **Web Access (HTTP)** remained enabled for compatibility with the installed legacy equipment. Their use on a production network requires review against the site's operational technology security policy.

6 I/O Scanning and PLC memory map

Here the TeSys T is the Modbus TCP server. The ETY4103 acts as the client/scanner: it periodically polls the controller and copies received values into PLC memory words.

Parameter	Status read	Current read
Server address	192.168.8.54	192.168.8.54
Unit ID	1	1
Read master object	%MW100	%MW104
Slave register reference	2502	500
Read length	4 words	8 words
Write length	0	0
Health timeout / repetitive rate	1500 ms / 500 ms	As configured for the read entry

6.1 Status registers

TeSys T register	PLC word	Status function
2502	%MW100	Mirror of register 455, system status 1
2503	%MW101	Mirror of register 456, system status 2
2504	%MW102	Mirror of register 457, logic inputs
2505	%MW103	Mirror of register 458, logic outputs

These words contain bit encoded states. They do not represent motor current.

6.2 Current registers

TeSys T registers	PLC words	Value	Scale stated
500–501	%MW104–%MW105	Average current (UDInt)	× 0.01 A
502–503	%MW106–%MW107	L1 current (UDInt)	× 0.01 A
504–505	%MW108–%MW109	L2 current (UDInt)	× 0.01 A
506–507	%MW110–%MW111	L3 current (UDInt)	× 0.01 A

The motor used in this test was single phase, so average current and L1 current were the relevant fields. The displayed current used low word %MW105. Before adapting this mapping to another installation, the exact word order and 32-bit handling should be checked against the applicable TeSys T Modbus manual.

IP address	Device Name	Unit ID	Slave Syntax	Health Timeout (ms)	Repetitive rate (ms)	RD Master Object	RD Ref Slave	RD length	Last value (Input)	WR Master Object	WR Ref Slave	WR length	Gateway/Bridge Device	Description
192.168.8.54		1	Index	1500	500	%MW100	2502	4	Set to 0	%MW4	0	0		
192.168.8.54		1	Index	1500	500	%MW104	500	8	Set to 0	%MW4	0	0		

Figure 5: I/O Scanning configuration

6.3 Additional registers mentioned

I also identified registers for a possible extension: %MW112 (thermal capacity, register 465), %MW113 (average current in % FLC, 466), %MW114 (L1 current in % FLC, 467), %MW119 (internal temperature, 472), %MW121 (frequency, 474), %MW128 (power factor, 481) and %MW129 (active power, 482). Their individual values would require validation before operational use.

7 Build and transfer

After configuring the hardware and network, I ran **Build > Rebuild All Project** and checked the result before transfer.

The build completed with **0 errors and 1 warning**. I reviewed the warning in relation to the services supported by the ETY4103 firmware; it was not a program syntax error.

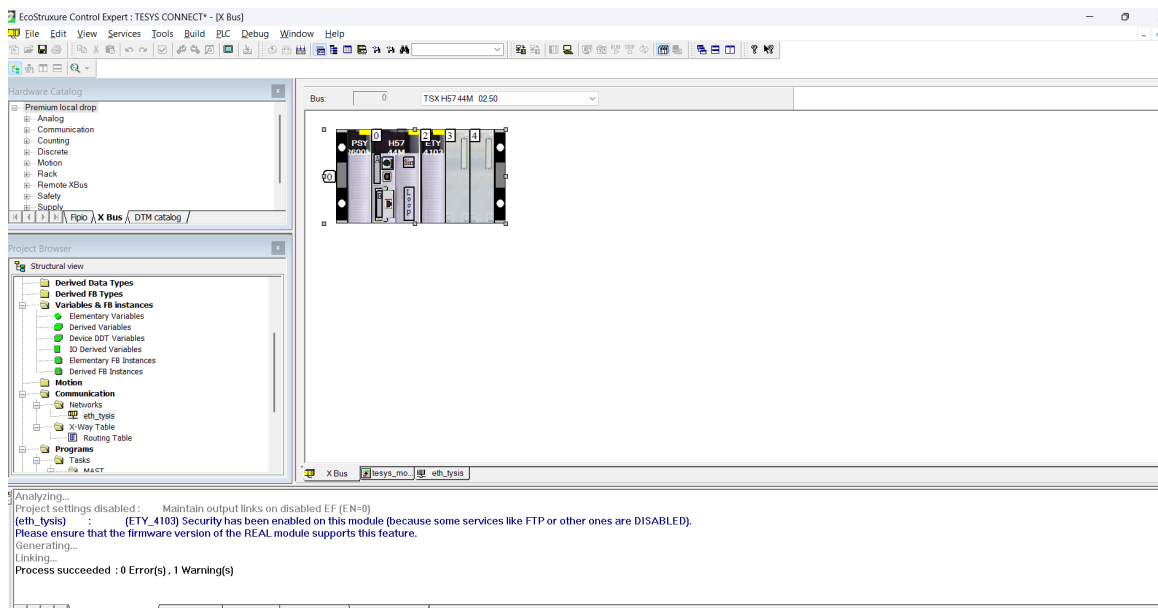


Figure 6: Project build result

I used the **SYS** system address over USB for the programming connection and selected the **PC-to-PLC** transfer direction to download the configured project. In Control Expert, a **PLC-to-PC** upload is a separate operation and depends on upload information being present in the PLC.

Before any transfer or switch to RUN, the motor, outputs and power circuit must be in a safe state for the test.

8 Communication validation

I checked connectivity in three stages: reaching the ETY4103 IP address, reaching the TeSys T IP address and testing TCP port 502.

Test	Target	Result obtained	What it verifies
ping 192.168.8.21	ETY4103	Replies received, no packet loss indicated	The device responds over IP
ping 192.168.8.54	TeSys T	Replies received, no packet loss indicated	The device responds over IP
Test-NetConnection 192.168.8.54 -Port 502	TeSys T Modbus TCP service	TcpTestSucceeded = True	TCP connections to port 502 are accepted

```

C:\Users\Cyborg>ping 192.168.8.21

Envoi d'une requête 'Ping' 192.168.8.21 avec 32 octets de données :
Réponse de 192.168.8.21 : octets=32 temps=2 ms TTL=64
Réponse de 192.168.8.21 : octets=32 temps=1 ms TTL=64
Réponse de 192.168.8.21 : octets=32 temps=1 ms TTL=64
Réponse de 192.168.8.21 : octets=32 temps=1 ms TTL=64

Statistiques Ping pour 192.168.8.21:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 1ms, Maximum = 2ms, Moyenne = 1ms

C:\Users\Cyborg>ping 192.168.8.54

Envoi d'une requête 'Ping' 192.168.8.54 avec 32 octets de données :
Réponse de 192.168.8.54 : octets=32 temps=3 ms TTL=255
Réponse de 192.168.8.54 : octets=32 temps=3 ms TTL=255
Réponse de 192.168.8.54 : octets=32 temps=1 ms TTL=255
Réponse de 192.168.8.54 : octets=32 temps=1 ms TTL=255

Statistiques Ping pour 192.168.8.54:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 1ms, Maximum = 3ms, Moyenne = 2ms

C:\Users\Cyborg>

```

ChatGPT can make mistakes. Check important info.

Figure 7: Ping checks for both devices

```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Cyborg> Test-NetConnection 192.168.8.54 -Port 502

ComputerName      : 192.168.8.54
RemoteAddress     : 192.168.8.54
RemotePort        : 502
InterfaceAlias    : Ethernet
SourceAddress     : 192.168.8.12
TcpTestSucceeded  : True

PS C:\Users\Cyborg>

```

Figure 8: Modbus TCP port check

A successful ping establishes IP reachability but does not establish that the registers are interpreted correctly. The port check confirms TCP availability; validating the data also requires checking the I/O Scanner, register alignment, scaling and online display.

9 Variables and supervision

The PLC variable table maps %MW100-%MW103 to four status words. %MW105 is the low word associated with `tesys_current`. I also defined variables for thermal capacity, internal temperature, frequency, power factor and active power.

Variable	Type	Address	Function
TeSys_Word_2502	WORD	%MW100	System status 1
TeSys_Word_2503	WORD	%MW101	System status 2
TeSys_Word_2504	WORD	%MW102	Logic inputs
TeSys_Word_2505	WORD	%MW103	Logic outputs
tesys_current	INT	%MW105	Low word of average current
TeSys_Current_A	REAL	—	Calculated value in amperes
Thermal_capacity_raw	INT	%MW112	Raw thermal capacity
Temperature_raw	INT	%MW119	Raw internal temperature
Frequency_raw	INT	%MW121	Raw frequency
PowerFactor_raw	INT	%MW128	Raw power factor
ActivePower_raw	INT	%MW129	Raw active power

These assignments provide named values for online monitoring and later integration into supervision. The status words must be decoded using the manufacturer's bit definitions before individual bits are used for alarms or control logic.

10 Results and measurement limitations

Online monitoring displayed an average current of **0.13 A** and an internal controller temperature of approximately **44 °C**. I mapped status and measurement registers into PLC memory, and verified IP connectivity and access to the Modbus TCP port.

The LTMR08EFM nominal measurement range is **0.4–8 A**. The displayed **0.13 A** was below that range. As a result:

- 0.13 A is an observed display result, not a demonstrated accurate measurement;
- zero or non-significant power factor or active power values at such a low current do not establish their normal measurement behavior;
- metrological validation would require a current within the specified range, an appropriate load, comparison with a reference instrument and confirmation of the single-phase wiring.

11 Commissioning and troubleshooting

Symptom or message	Check or corrective action
Compatibility check fails during transfer	Check the CPU reference and project version (02.50 in this case), then the ETY4103 reference and enabled options
0 Error(s), 1 Warning(s)	The project built without errors; review the warning against the firmware and enabled security services
Ping fails	Check unique IP addresses, the /24 mask, cables and that devices connect to the same switch
Ping succeeds, Modbus communication remains unverified	Check TCP port 502, Unit ID 1, server IP and I/O Scanner rows
PLC-to-PC upload unavailable	Check whether upload information is present; distinguish upload from downloading a project to the PLC

Symptom or message	Check or corrective action
Zero or non-significant current/power	Check whether load current is within the TeSys T measurement range and that scaling and word order match the manual

12 Operational and cybersecurity considerations

I configured both I/O Scanning rows with `WR length = 0`, so the Modbus exchanges implemented here read registers and do not send motor commands. This configuration does not replace checks of the PLC outputs or the machine's safe operating state.

FTP/TFTP and HTTP services are described as enabled for compatibility on the test network. Before connection to a production network, the site's OT/security team should review these services and disable or isolate them as appropriate. A project transfer and switch to RUN should take place only after the machine's safe state has been verified.

13 Conclusion and next steps

I configured the TSX H57 44M and TeSys T LTMR08EFM communication path: the rack and CPU version matched the installed hardware, the ETY4103 used a defined IP address, two Modbus TCP read areas were mapped into `%MW` words, both devices answered ping requests and the TeSys T accepted a TCP connection on port 502.

Next steps are end-to-end validation with motor load current inside the measurement range, confirmation of word order and scaling against the TeSys T manual, and archiving the full ST listing and complete animation table. Bit-level decoding of status words 2502–2505 should follow the applicable manufacturer's definitions before operational use.

14 Technical reference

Schneider Electric, **TeSys T Communication Variables**, document DOCA0130FR-03 (register and bit definitions).